

UAS OneSpan SEE with nShield Connect Integration Guide - 6.0.1-GA

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1. Preface - UAS OneSpan SEE with nShield Connect Integration Guide

This guide contains instructions on how to integrate OneSpan (formerly VASCO) Entrust Secure Execution Engine (SEE) with Entrust nCipher Connect HSM.

Intended Reader

This document is intended as reference for administrators.

Document Scope

This guide provides instructions on configuring the AccessMatrix Universal Authentication Server (UAS) to support the integration of OneSpan (formerly VASCO) SEE with the Entrust nCipher Connect HSM.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix UAS (Universal Authentication Server) documentation set:

- [*AccessMatrix Common Administration Guide*](#)
- [*UAS Administration Guide*](#)
- [*UAS Entrust nShield Connect HSM Deployment Guide*](#)

2. Introduction to UAS OneSpan SEE with nShield Connect Integration

This user guide presents detailed explanations and steps on how the AccessMatrix Universal Authentication Server (UAS) can be configured to provide support for OneSpan SEE integration with a Entrust nCipher Connect HSM.

Prerequisite

1. Ensure that client machines and RFS machine are configured.
2. Prepare a privileged client machine to load the SEE Machine.
3. Make sure that KeySafe can be started up in the privileged client machine.
4. Install OneSpan VACMAN Controller nCipher HSM 3.14.0 in the privileged client machine.

SEE Machine Deployment Flow

Please refer to the following sections for SEE machine deployment:

- [Enabling SEE Feature \(a.k.a SEE License\)](#)
- [Generating Keys](#) (SEE Signing Key and Userdata Signing Key)
- [Requesting SEE Dynamic Certificate Feature - ADDER Certificate](#) (optional, only for Restricted SEE module)
- [Signing and Packing](#) (SEE machine and Userdata)
- [Auto Loading SEE Machine](#)

Integrating OneSpan SEE with Entrust nCipher Connect

Please refer to the following sections for integrating OneSpan SEE:

- [Managing OneSpan Keys](#)
 - [Configuring Admin Console](#)
-

3. OneSpan SEE Integration

The following sections provide information on carrying out the OneSpan SEE integration with a Entrust nCipher Connect HSM:

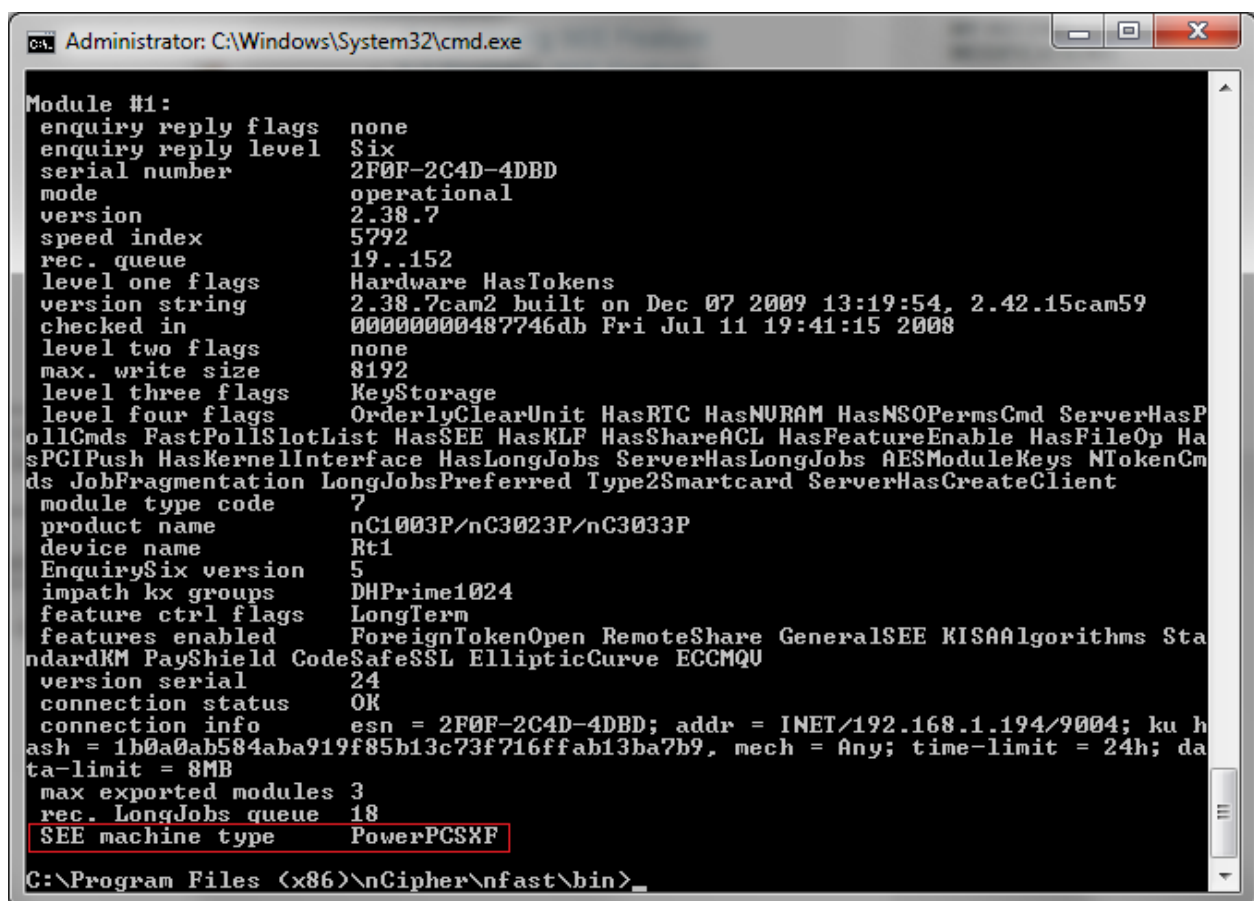
- [Deploying SEE Machine](#)
- [Integrating OneSpan SEE with Entrust nCipher Connect](#)
- [Configuring Admin Console](#)

3.1. Deploying SEE Machine

The Secure Execution Engine (SEE) is a feature available on some Entrust HSMs that allows the application code to be executed within the security boundary of the module.

The SEE is a unique secure execution environment. It is an executable binary code that runs within the security boundary of the module, cross compiled to native code on the host. It can be signed, encrypted or both with the Trusted Code Tool (tct2).

HSM modules run either PowerPC-based or ARM-based binaries, depending on module type (the use of ARM-based binaries is deprecated). To determine your module type, run the `enquiry` command-line utility and check the SEE Machine Type output. The image below is an example of the output:



```
Administrator: C:\Windows\System32\cmd.exe

Module #1:
enquiry reply flags      none
enquiry reply level     Six
serial number           2F0F-2C4D-4DBD
mode                    operational
version                 2.38.7
speed index             5792
rec. queue              19..152
level one flags         Hardware HasTokens
version string          2.38.7cam2 built on Dec 07 2009 13:19:54, 2.42.15cam59
checked in              00000000487746db Fri Jul 11 19:41:15 2008
level two flags         none
max. write size         8192
level three flags       KeyStorage
level four flags        OrderlyClearUnit HasRTC HasNVRAM HasNSOPermsCmd ServerHasP
ollCmds FastPollSlotList HasSEE HasKLF HasShareACL HasFeatureEnable HasFileOp Ha
sPCIPush HasKernelInterface HasLongJobs ServerHasLongJobs AESModuleKeys NTokenCm
ds JobFragmentation LongJobsPreferred Type2Smartcard ServerHasCreateClient
module type code        7
product name            nC1003P/nC3023P/nC3033P
device name             Rt1
EnquirySix version      5
impath kx groups        DHPrime1024
feature ctrl flags      LongTerm
features enabled         ForeignTokenOpen RemoteShare GeneralSEE KISAAlgorithms Sta
ndardKM PayShield CodeSafeSSL EllipticCurve ECCMQU
version serial          24
connection status        OK
connection info          esn = 2F0F-2C4D-4DBD; addr = INET/192.168.1.194/9004; ku h
ash = 1b0a0ab584aba919f85b13c73f716ffab13ba7b9, mech = Any; time-limit = 24h; da
ta-limit = 8MB
max exported modules    3
rec. LongJobs queue     18
SEE machine type        PowerPCSKF

C:\Program Files (x86)\nCipher\nfast\bin>
```

Enabling SEE Feature (a.k.a SEE License)

Note: It is assumed that you have the SEE Feature enabling certificate prior to performing steps below.

From module's front panel:

1. **Enabling SEE feature with a smart card:**

- a. Insert the Feature Enabling smart card into the module's slot.
- b. Read the FEM from the card (from menu 2-3-1).
- c. A message is displayed if the features are successfully enabled.

2. **Enabling SEE feature without a smart card:**

- a. Put the file that contains the SEE feature enabling certificate in `%NFAST_KMDATA%\<hsm-ESN>\features\` on the remote file system.
- b. You can give the file any appropriate name. You must enter the file name on the module front panel.
- c. Read from a file (from menu 2-3-2).
- d. Specify the name of the file that contains the certificate.
- e. If the SEE feature is successfully enabled, a message confirms this.

From client's command prompt:

1. Run the command prompt as an administrator in the RFS machine and navigate to `%NFAST_HOME%\bin\`.
2. Run the Feature Enable Tool to enable the SEE feature with the command `fet`.

3. **Enabling SEE feature with a smart card:**

- a. Enter "1" to read FEM certificate(s) from a smart card or cards.
- b. Press Enter when you are prompted to insert FEM card(s) into the module slot(s).
- c. A message is displayed if the features are successfully enabled.

4. **Enabling SEE feature without a smart card:**

- a. Put the file that contains the SEE feature enabling certificate in `%NFAST_KMDATA%\<hsm-ESN>\features\` on the remote file system.
 - b. You can give the file any appropriate name. You must enter the file name on the module front panel.
 - c. Read from a file (from menu 2-3-2).
 - d. Specify the name of the file that contains the certificate.
 - e. If the SEE feature is successfully enabled, a message confirms this.
-

Viewing Enabled Features

To view the status of or verify features on an nShield Connect unit, use the front panel or the `fet` utility.

From module's front panel:

1. To see which features have already been enabled on the nShield Connect, access menu 1-2-3-3.

From privileged client's command prompt:

1. Run command prompt as an administrator in the client machine and navigate to `%NFAST_HOME%\bin\`.
2. View enabled features by running the command `fet`.

Generating Keys

Whenever possible, generate a new key instead of importing an existing key. If existing keys are stored in a known format on your hard disk, there is a risk that the existing key has been compromised.

- Generate a key from a privileged client's KeySafe, or via the command prompt.

Generating SEE Code Signing Key

i	Note: There are two ways to generate SEE Code Signing Key: 1. KeySafe 2. Command Prompt
----------	--

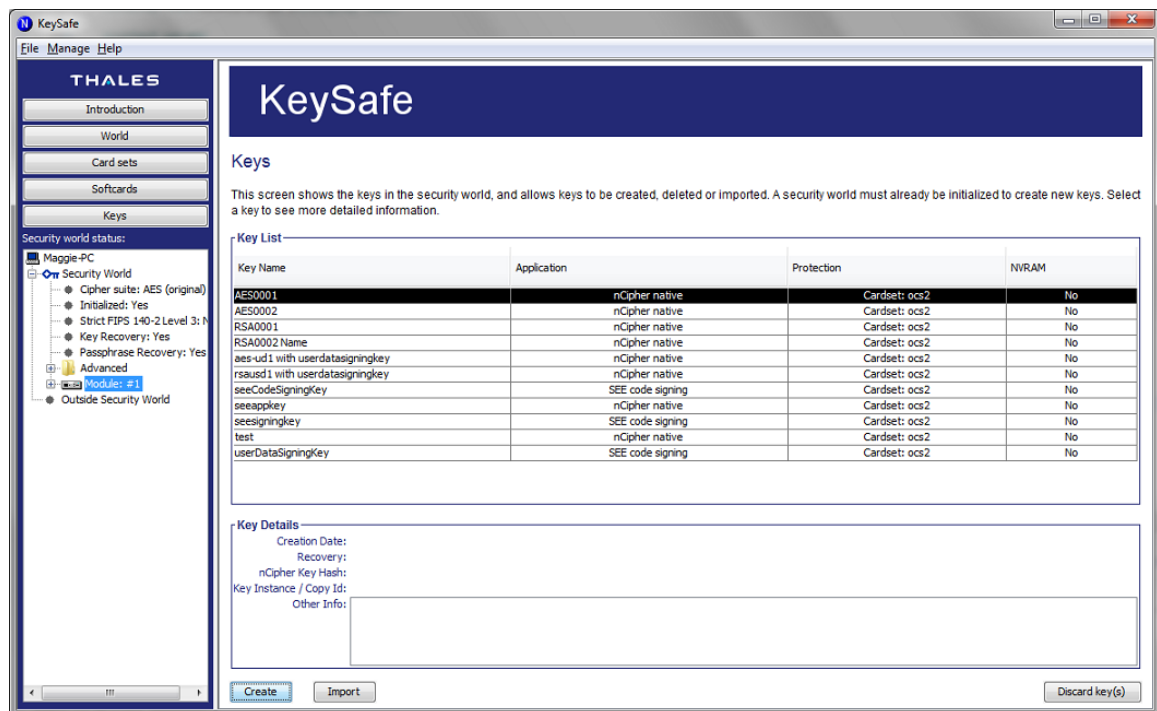
From KeySafe:

1. Start KeySafe by selecting the KeySafe entry from the Start menu under **All Programs > nCipher**.

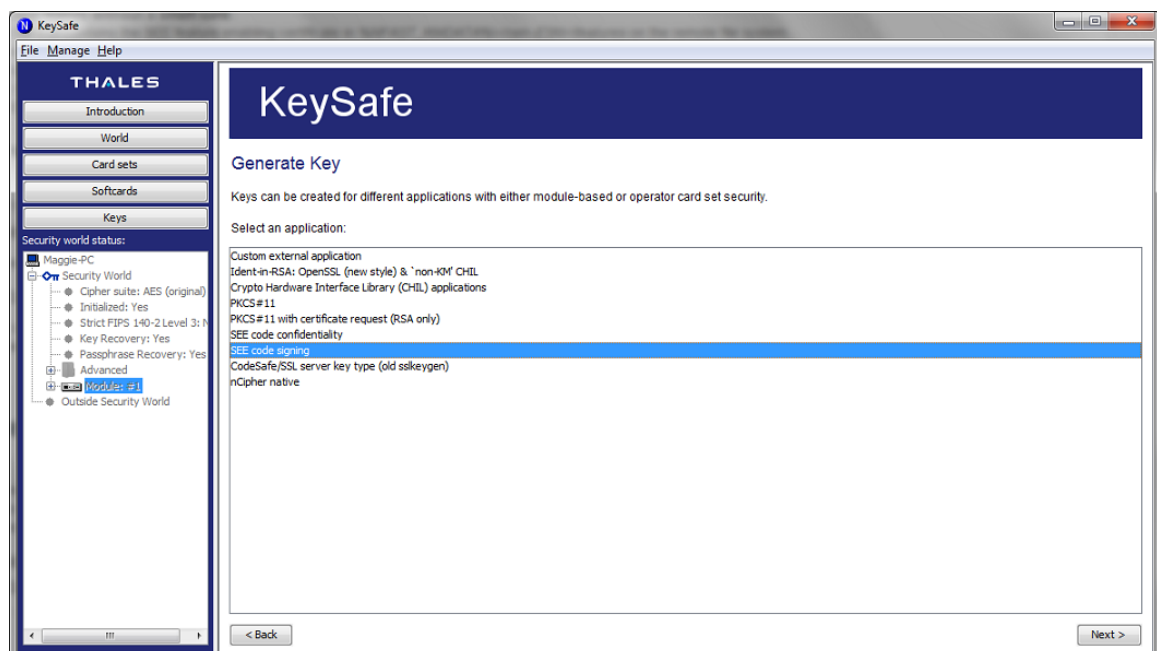
i	Note: Administrator privilege is needed.
----------	---

2. When it starts, click on the **Keys** button on the sidebar as shown in the image below.
-

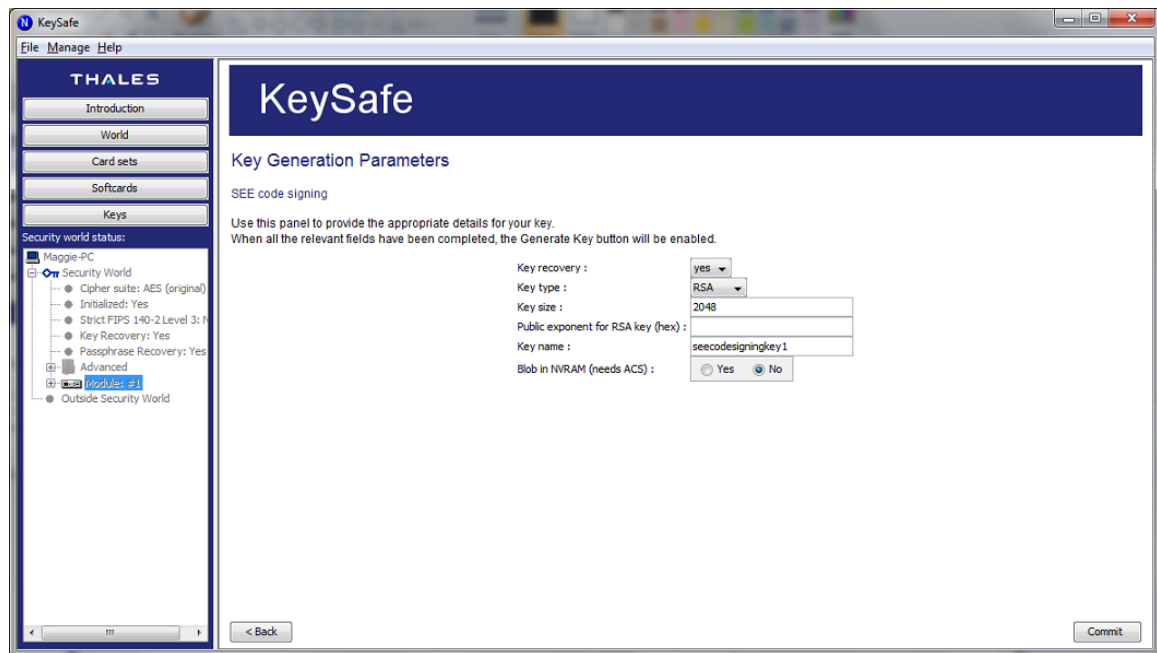
- Click on the **Create** button at the bottom of the page to generate a new key.



- Select the **SEE code signing** application:



5. Enter "2048" for **Key size** and an appropriate **Key name**, while leaving the other attributes as their default values:

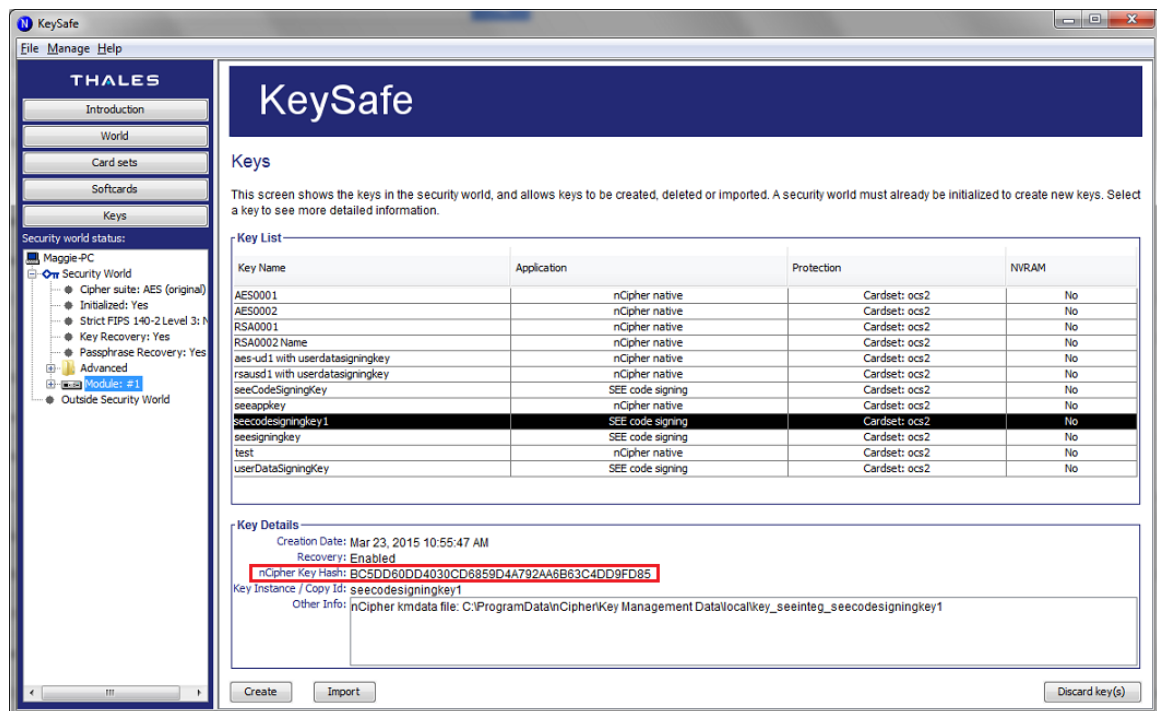


6. Fill in the parameters and click on the **Commit** button. It will prompt a success message box, as shown below:



7. Close the message box and click on the **Keys** button on the sidebar, which takes you back to the **Key Listing** page to view the new generated key.

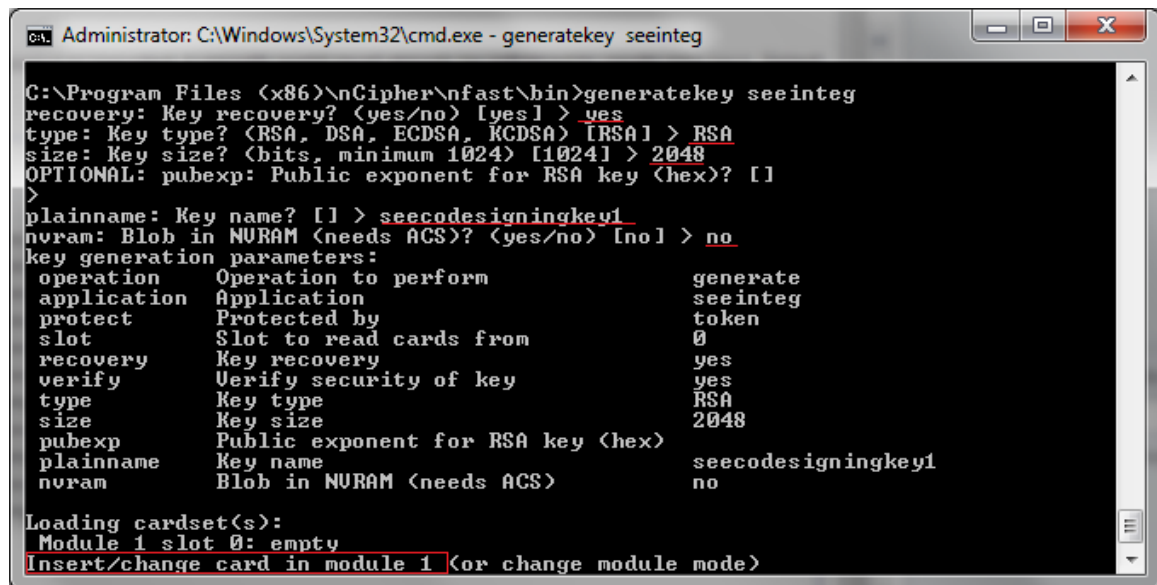
8. The **nCipher Key Hash** value for the SEE Code Signing Key is important for code signing later:



From Command Prompt:

1. Run the command prompt as an administrator in the client machine and navigate to `%NFAST_HOME%\bin\`.
2. Generate a key for signing the SEE machine by running the command `generatekey seeinteg`. After running the command, it will prompt you to key in the required parameters.
3. Supply the following values:
 - a. Key recovery? - Enter "yes".
 - b. Key type? - Enter "RSA".
 - c. Key size? - Enter "2048".
 - d. Public exponent for RSA key (hex)? - Leave this blank.
 - e. Key name? - Enter an appropriate name.
 - f. Blob in NVRAM (need ACS)? - Enter "no".

4. After completing the last parameter, press Enter. A detail of key generation parameters will be displayed as shown in image below. It will also prompt you to insert the cardset:

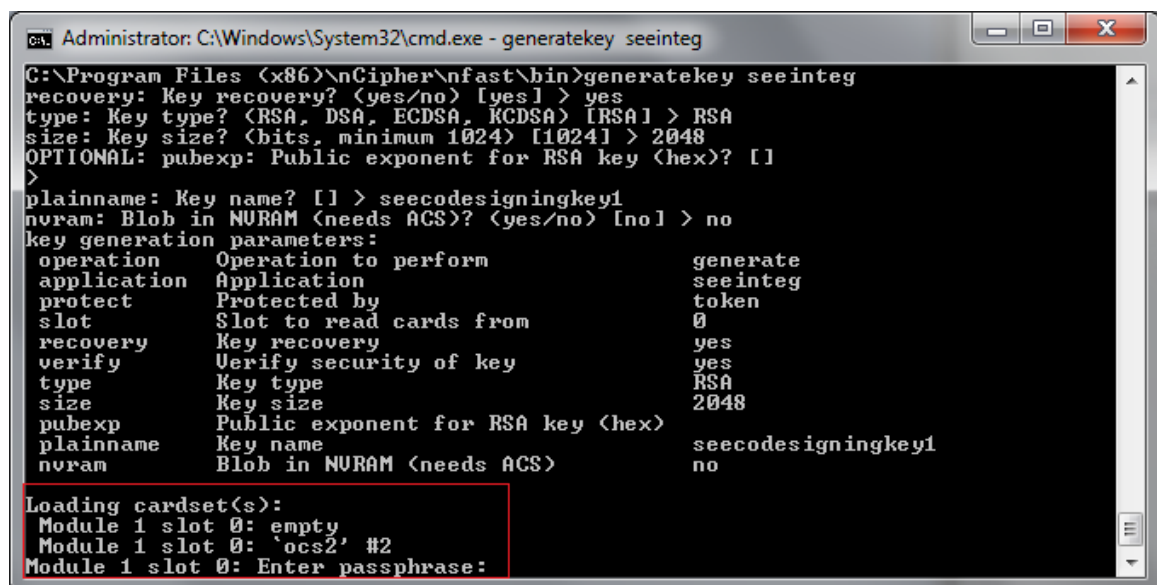


```
Administrator: C:\Windows\System32\cmd.exe - generatekey seeinteg

C:\Program Files (x86)\nCipher\nfast\bin>generatekey seeinteg
recovery: Key recovery? (yes/no) [yes] > yes
type: Key type? (RSA, DSA, ECDSA, KCDSA) [RSA] > RSA
size: Key size? (bits, minimum 1024) [1024] > 2048
OPTIONAL: pubexp: Public exponent for RSA key (hex)? []
>
plainname: Key name? [] > seecodesigningkey1
nvrnm: Blob in NURAM (needs ACS)? (yes/no) [no] > no
key generation parameters:
  operation      Operation to perform      generate
  application    Application                      seeinteg
  protect        Protected by                       token
  slot           Slot to read cards from           0
  recovery       Key recovery                      yes
  verify         Verify security of key           yes
  type           Key type                         RSA
  size           Key size                        2048
  pubexp         Public exponent for RSA key (hex)
  plainname      Key name                        seecodesigningkey1
  nvrnm          Blob in NURAM (needs ACS)        no

Loading cardset(s):
Module 1 slot 0: empty
Insert/change card in module 1 (or change module mode)
```

5. Insert the OCS card into card slot at module front panel. You are prompted to enter the passphrase of the card - do so:

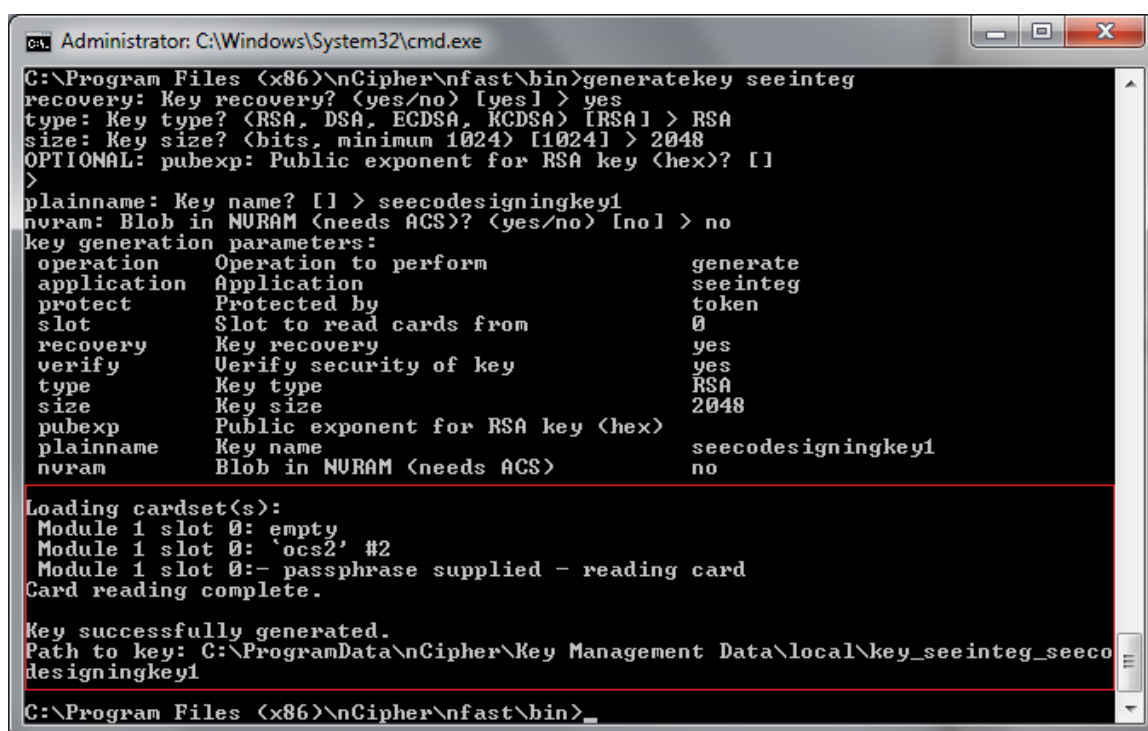


```
Administrator: C:\Windows\System32\cmd.exe - generatekey seeinteg

C:\Program Files (x86)\nCipher\nfast\bin>generatekey seeinteg
recovery: Key recovery? (yes/no) [yes] > yes
type: Key type? (RSA, DSA, ECDSA, KCDSA) [RSA] > RSA
size: Key size? (bits, minimum 1024) [1024] > 2048
OPTIONAL: pubexp: Public exponent for RSA key (hex)? []
>
plainname: Key name? [] > seecodesigningkey1
nvrnm: Blob in NURAM (needs ACS)? (yes/no) [no] > no
key generation parameters:
  operation      Operation to perform      generate
  application    Application                      seeinteg
  protect        Protected by                       token
  slot           Slot to read cards from           0
  recovery       Key recovery                      yes
  verify         Verify security of key           yes
  type           Key type                         RSA
  size           Key size                        2048
  pubexp         Public exponent for RSA key (hex)
  plainname      Key name                        seecodesigningkey1
  nvrnm          Blob in NURAM (needs ACS)        no

Loading cardset(s):
Module 1 slot 0: empty
Module 1 slot 0: 'ocs2' #2
Module 1 slot 0: Enter passphrase:
```

6. A Key successfully generated message is displayed after completion, as shown below:



```
Administrator: C:\Windows\System32\cmd.exe
C:\Program Files (x86)\nCipher\nfast\bin>generatekey seeinteg
recovery: Key recovery? (yes/no) [yes] > yes
type: Key type? (RSA, DSA, ECDSA, KCDSA) [RSA] > RSA
size: Key size? (bits, minimum 1024) [1024] > 2048
OPTIONAL: pubexp: Public exponent for RSA key (hex)? []
>
plainname: Key name? [] > seecodesigningkey1
nvram: Blob in NVRAM (needs ACS)? (yes/no) [no] > no
key generation parameters:
operation      Operation to perform          generate
application    Application                      seeinteg
protect        Protected by                       token
slot           Slot to read cards from           0
recovery       Key recovery                      yes
verify         Verify security of key           yes
type           Key type                         RSA
size           Key size                        2048
pubexp         Public exponent for RSA key (hex)
plainname      Key name                        seecodesigningkey1
nvram          Blob in NVRAM (needs ACS)         no

Loading cardset(s):
Module 1 slot 0: empty
Module 1 slot 0: 'ocs2' #2
Module 1 slot 0:- passphrase supplied - reading card
Card reading complete.

Key successfully generated.
Path to key: C:\ProgramData\nCipher\Key Management Data\local\key_seeinteg_seeco
designingkey1
C:\Program Files (x86)\nCipher\nfast\bin>
```

Generating Userdata Signing Key

1. The procedure to generate the Userdata Signing Key is the same as the procedure for generating the SEE Code Signing Key. Repeat the process from [Generating SEE Code Signing Key](#).

Requesting SEE Dynamic Certificate Feature - ADDER Certificate

The SEE is a unique secure execution environment. The available SEE features are:

SEE Features	Description
SEE Activation (EU+10)	This SEE feature is provided with the CodeSafe developer product to enable you to develop and run SEE applications. The CodeSafe developer product is only available to customers in the Community General Export Area (CGEA, also known as EU+10). Contact Entrust to find out whether your country is currently within the CGEA. The customer uses their own SEEcode signing key and userdata signing key. ADDER cert is NOT required.
SEE Activation (Restricted)	SEE machines must be signed with a key "blessed" by Entrust to be loaded on the HSM, it requires ADDER certificate issued by Entrust. The customer uses their own SEEcode signing key and userdata signing key. ADDER cert is required.



Notes:

- To deploy an already-developed SEE machine, you require the dynamic Feature Enabling certificate chain.
- Customers who require the dynamic certificate chain can load an SEE machine only when the key used to sign the SEE machine is export-approved by Entrust through the provision of a signing certificate (an ADDER certificate).
- Dynamic certificate chain is valid only on the Restricted SEE module.

Obtaining ADDER Certificate

1. Send the SEE code signing key hash to Entrust together with a description of the SEE machine by email.
2. Entrust approves the SEE machine for export and sends you an ADDER certificate to allow the SEE machine signed by the specified key to run.

Using ADDER Certificate

1. Develop an installation process that places the certificate in the `%NFAST_KMDATA%\Feature Certificates\` folder.

Signing and Packing

Before signing and packing SEE Machine, ensure the following:

- During the KeySafe key-generation process, select the **SEE Code Integrity** option.
- When generating the key with the `generatekey` command-line utility, select the application type `seeinteg`.

Sign and pack the SEE machine via the privileged client's command prompt.

Signing and Packing SEE Machine

1. Copy the SXF (Secure eXecution Format) file **aal2sdk.sxf** from `<Vasco installed path>\VACMAN Controller nCipher HSM 3.14.0\Hsm\ppc\` into `%NFAST_HOME%\bin\`.



Note: If Vasco Multi-Device token vendor is used, please use `<Vasco installed path>\VACMAN Controller nCipher HSM 3.15.0\Hsm\ppc\` instead.

-
2. Use the generated SEE Code Signing Key to sign and pack the SXF into a SAR (Secure or SEE ARchive) file by running the command:

```
tct2 --sign-and-pack --is-machine -i aal2sdk.sxf -o <outfile sar file name> -k  
<SEE code signing key name>
```

3. After running the command, an output file with the .sar file extension will appear in
`%NFAST_HOME%\bin\`.

Signing and Packing User Data

i **Note:** In the current release of software, all SEE machines built with bsdlb must be provided with a valid ASCII-format CPIO archive. This archive forms the base of the file system available to your SEE machine. Even if your SEE machine does not use this file system, you must still create and supply it with dummy userdata as a place-holder.

1. Create a dummy userdata file suitable for loading into the module (i.e. a text file named **userdata.txt** containing the word `vascoSEE` in `%NFAST_HOME%\bin\`).
2. `vascoSEE` is the name of published object that is used to create a hostside SEE World later.
3. The SEE code is supposed to be signed by the vendor of SEE machine, but sometimes this is not the case. Therefore, the customer needs to create a SEE Code Signing Key and apply an ADDER certificate from Entrust if customer is using HSM with restricted SEE license. This ADDER certificate need to be placed inside Dynamic Feature Certificates folder of RFS.
4. Turn it into a SAR file with the `tct2` command-line utility:

a. SEE Signing Key and Userdata Signing Key in same Security World

- i. Sign the user data using the customer user data signing key + SEE code signing key.
- ii. Copy the ADDER certificate to the Feature Certificates folder of RFS:

```
tct2 --sign-and-pack --machine-key-ident <SEE code signing key name>  
-o <outfile sar file name> -i userdata.txt -k <userdata signing key  
name>
```

i **Note:** The `<outfile sar file name>` of the userdata must be named **userdata.sar**, else VASCO Manager Tool will not able to read it.

b. SEE Signing Key and Userdata Signing Key in different Security Worlds

- i. Sign the user data using the customer user data signing key + key hash of SEE code signing key.
-

-
- ii. Copy the ADDER certificate to the Feature Certificates folder of RFS:

```
tct2 --sign-and-pack --machine-key=<SEE code signing key's Key Hash>
-o <outfile sar file name> -i userdata.txt -k <userdata signing key
name>
```

Auto Loading SEE Machine



Note: Configure auto load and publish SEE Machine in privileged client machine.

Configuring Auto Load SEE Machine

To configure an nShield Connect module to autonomously run an SEE machine and accept updates using the RFS:

1. Copy the existing config file from `%NFAST_KMDATA%\Key Management Data\config\` to `%NFAST_HOME%\bin\` with a new file named **config.new**.
2. In the `load_seemachine` section of the **config.new** file for the remote module, add or amend the following settings:

Code

```
pull_rfs = true
machine_file = mymachinename.sar
userdata = myuserdata.sar
worldid_pubname = publ_name
```



Note: `worldid_pubname` is the dummy data in **myuserdata.sar** file.

3. In the `sys_log` section of the **config.new** file for the remote module, add or amend the following settings:

Code

```
behaviour = push
push_interval = 1
```

4. Load the new updated configuration file on to the module. To do this, run the following command:

```
cfg-pushnethsm --address=<module_IP_address> --force config.new
```



Note: These settings control how and where log messages are written. Using the example above, messages will be written to the **system.log** and **hardserver.log** files of the module, which are accessible using the remote file system. You may wish to revise `push_interval` to a higher value once the nShield Connect has successfully loaded the new SEE machine.

5. Run `nopclearfail` to clear the existing SEE machine in one of the modules or ALL the modules.

```
nopclearfail --clear --module=<module number>
```

OR

```
nopclearfail --clear --all
```

Loading and Publishing SEE Machine

The `hsc_loadseemachine` utility enables you to publish the SEE machine. The utility:

- Loads the SEE machine into each module configured.
- Publishes a newly created SEE world, if appropriate.

To do so, run the following command:

```
hsc_loadseemachine --module=<module number> --configfile=<new Config FileName>
```

3.2. Integrating VASCO SEE with Thales nCipher Connect

Fulfill the following prerequisites to integrate OneSpan SEE with Entrust nCipher Connect HSM:

1. Ensure that the client machine is installed with JAVA 1.5x or latest version.
2. Ensure that the client machine is installed with OneSpan VACMAN Controller.
3. Ensure that the production version of nCipher Software - nCSS is installed in the client machine.
4. Ensure that the AccessMatrix Server is configured running properly.
5. Copy the following libraries and jar files from the OneSpan installation folder to AccessMatrix installation folder:
 - a. **aal2wrap.jar** - from *<OneSpan installed path>\Wrappers\Java* to *tomcat\webapps\am5\WEB-INF\lib*
 - b. **aal2sdk.dll** - from *<OneSpan installed path>\Bin* to *tomcat\bin*
6. Copy the following libraries and jar files from the nCipher installation folder to AccessMatrix installation folder:
 - a. **Nfjava.jar** - from *%NFAST_HOME%\java\classes* to *tomcat\webapps\am5\WEB-INF\lib*
 - b. **kmjava.jar** - from *%NFAST_HOME%\java\classes* to *tomcat\webapps\am5\WEB-INF\lib*
 - c. **jhsee.jar** - from *%NFAST_HOME%\java\classes* to *tomcat\webapps\am5\WEB-INF\lib*

Managing OneSpan Keys

OneSpan provides the customer with secrets in the form of a DPX file. The customer stores the secrets as a collection of DIGIPASS BLOBs. The secrets have to be protected at all times, including during the transport of the DPX file and storage of the DIGIPASS application BLOB. The protection of the DPX files and DIGIPASS application BLOBs is based on cryptographic operations with keys, which need to be protected as well. The HSM is used to safely manage the keys that are used to protect DPX files and DIGIPASS application BLOBs.

The cryptographic keys that are involved in the protection mechanisms:

Keys	Description
HSM-Level DPX Transport Key	The HSM-level DPX transport key is used to encrypt the sensitive DIGIPASS information in a DPX file. This key is typically generated by the customer and distributed to OneSpan, encrypted with a key encrypting key (KEK).
HSM-Level BLOB Storage Key	The HSM-level BLOB storage key is used to encrypt the sensitive DIGIPASS information of the DIGIPASS application BLOB for long-term storage.
Key Encrypting Key (KEK)	The key encrypting key (KEK) is typically used by the customer to deliver the HSM-level DPX transport key to OneSpan.

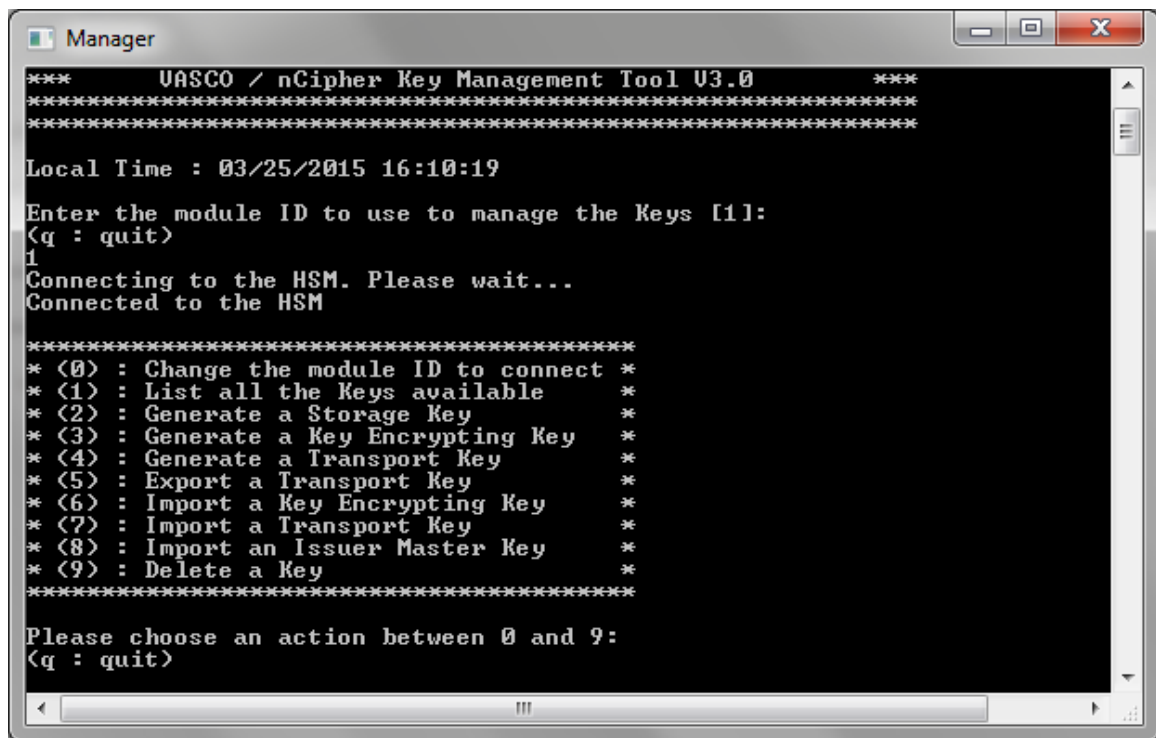
Generating and Importing Cryptographic Keys

i	Notes: - The name of the keys that are sent to OneSpan for DPX file generation must be the same as the keys that are generated by Key Manager Tool. - These keys are sensitive. Please refer to Appendix B: Customer's Key Attributes Permission for the key's attribute permission.
----------	---

Use the Key Management Tool to generate the keys. Locate and run **manager.exe** in *<OneSpan installed path>\Manager*. Before generating any key, please go through the following steps:

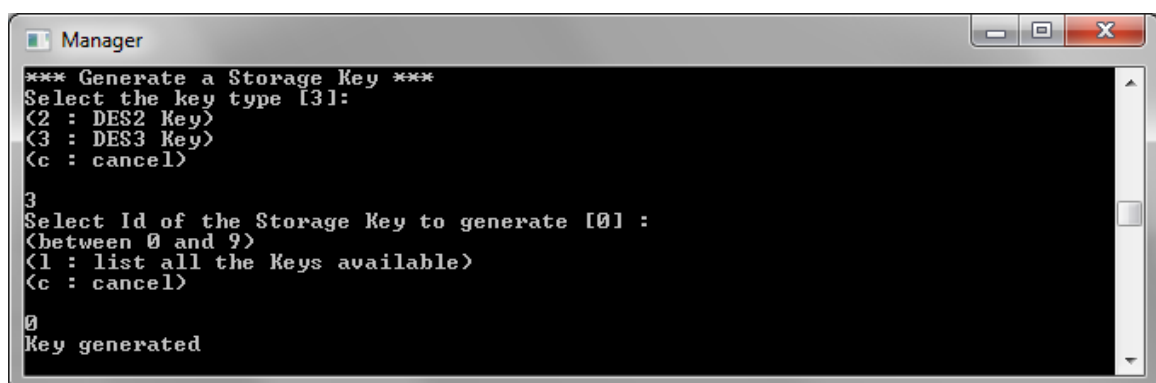
1. Ensure that the nCipher Software is installed in client and RFS machines and well connected with nShield Connect module. Run `enquiry` in the command prompt to see if the module is displayed.
2. Place the SEE machine and the Userdata that is already signed under the path `%NFAST_KMDATA%\Key Management Data\`.

3. Enter module ID as shown in image below to connect to HSM module:



Generating Storage Key

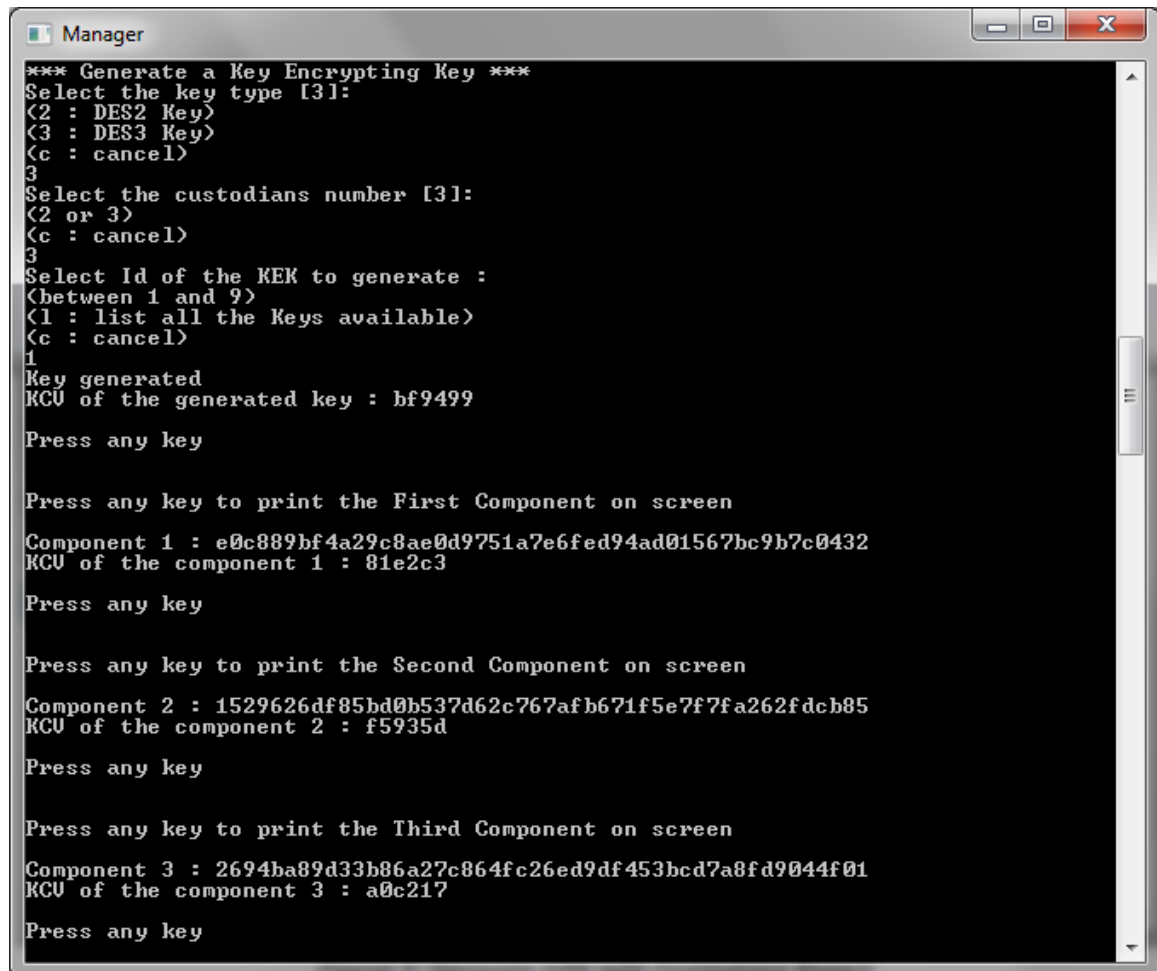
1. On the OneSpan / nCipher Key Management Tool main screen, select (2): Generate a Storage Key.
2. Select the key type DES2 or DES3. The default value is 3.
3. Select the ID of the storage key. The default value is 0.



Generating KEK with Custodians Export

1. On the OneSpan / nCipher Key Management Tool main screen, select (3): Generate a Key Encrypting Key.
2. Select the key type DES2 or DES3. The default value is 3.

3. Select the number of custodians. The default value is 3.
4. Select the ID of the KEK. Choose between 1 to 9 (in this example, choose "1").
5. Press any key to continue.
6. Each Hexadecimal component 1, 2, and 3, the value of the KEK and KCV of the custodians will be displayed on the screen.



```
*** Generate a Key Encrypting Key ***
Select the key type [3]:
<2 : DES2 Key>
<3 : DES3 Key>
<c : cancel>
3
Select the custodians number [3]:
<2 or 3>
<c : cancel>
3
Select Id of the KEK to generate :
<between 1 and 9>
<1 : list all the Keys available>
<c : cancel>
1
Key generated
KCV of the generated key : bf9499

Press any key

Press any key to print the First Component on screen
Component 1 : e0c889bf4a29c8ae0d9751a7e6fed94ad01567bc9b7c0432
KCV of the component 1 : 81e2c3

Press any key

Press any key to print the Second Component on screen
Component 2 : 1529626df85bd0b537d62c767afb671f5e7f7fa262fdcb85
KCV of the component 2 : f5935d

Press any key

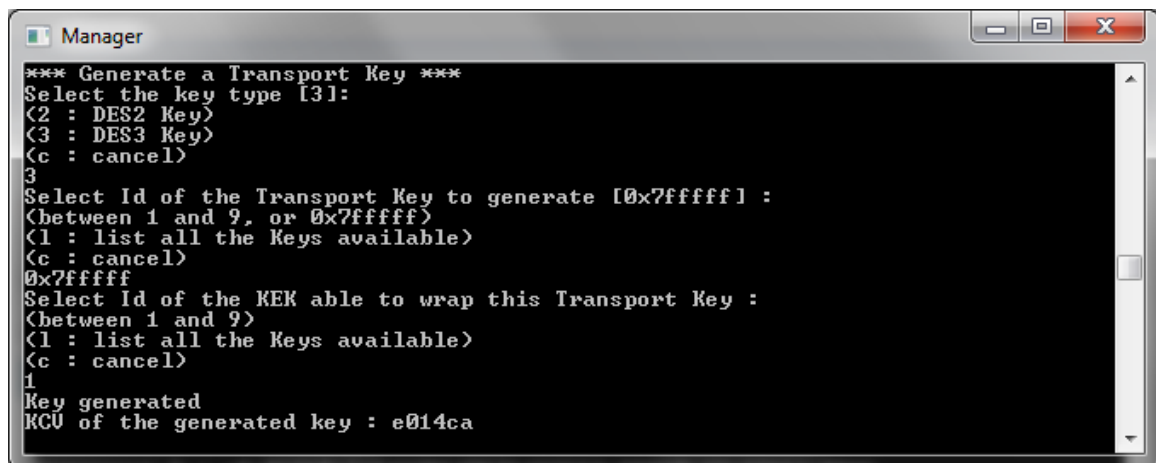
Press any key to print the Third Component on screen
Component 3 : 2694ba89d33b86a27c864fc26ed9df453bcd7a8fd9044f01
KCV of the component 3 : a0c217

Press any key
```

Generating Transport Key

1. On the OneSpan / nCipher Key Management Tool main screen, select (4): Generate a Transport Key.
2. Select the key type DES2 or DES3. The default value is 3.
3. Select the ID of the Transport Key. The default value is 0x7fffff.
4. Select the ID of the KEK (this key will be used to unwrap the wrapped Transport Key).

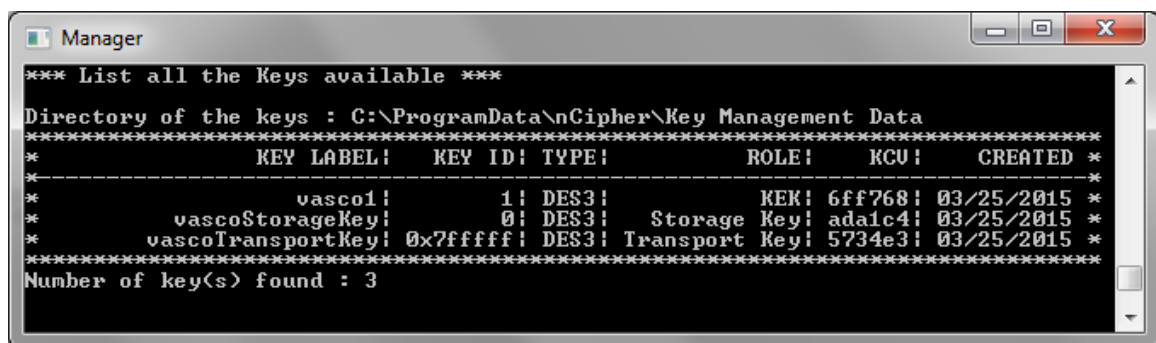
5. The KCV of the Transport Key will be displayed on the screen.



```
*** Generate a Transport Key ***
Select the key type [3]:
(2 : DES2 Key)
(3 : DES3 Key)
(c : cancel)
3
Select Id of the Transport Key to generate [0x7fffff] :
(between 1 and 9, or 0x7fffff)
(1 : list all the Keys available)
(c : cancel)
0x7fffff
Select Id of the KEK able to wrap this Transport Key :
(between 1 and 9)
(1 : list all the Keys available)
(c : cancel)
1
Key generated
KCV of the generated key : e014ca
```

Listing Available Keys

1. On the OneSpan / nCipher Key Management Tool main screen, select (1): List all the Keys available.



```
*** List all the Keys available ***
Directory of the keys : C:\ProgramData\nCipher\Key Management Data
*****
*          KEY LABEL!   KEY ID! TYPE!          ROLE!   KCV!   CREATED *
*****
*          vasco1!      1! DES3!          KEK!  6ff768! 03/25/2015 *
*      vascoStorageKey! 0! DES3!      Storage Key! ada1c4! 03/25/2015 *
*      vascoTransportKey! 0x7fffff! DES3! Transport Key! 5734e3! 03/25/2015 *
*****
Number of key(s) found : 3
```

2. 3 text files, **OneSpan<ID>.txt**, **OneSpanStorageKey.txt**, and **OneSpanTransportKey.txt** will be displayed in %NFAST_KMDATA%\Key Management Data\.

Deleting Key

1. On the OneSpan / nCipher Key Management Tool main screen, select (9): Delete a Key.
2. Select ID of the key to delete (in this example, choose "1" to delete KEK).
3. Selected key will display on the screen.

-
4. Key in "Y" or "N" to confirm deletion.



```
*** Delete a Key ***
Select Id of the key to delete :
(between 0 and 9, or 0x7ffff)
(l : list all the Keys available)
(c : cancel)
1
*****
*                KEY LABEL!  KEY ID! TYPE!                ROLE!  KCV!  CREATED *
*-----*
*                vasco1!      1! DES3!                KEK! 6ff768! 03/25/2015 *
*****
Are you sure to delete the key above? [N]
(Y or N)
(c : cancel)
y
Key deleted
```

3.3. Configuring Admin Console

Validating Token Vendor

To use OneSpan (formerly OneSpan) SEE token, AccessMatrix server requires its server license to be enabled with VASCO_FM_THALES token vendor. This vendor is required during OneSpan SEE token import.

To check the availability of VASCO_FM_THALES:

1. Perform the steps described in [Navigating Server License](#).
2. Look for **UAS FEATURE** under the **Licence Module** column and **Token Vendors** under the **Feature** column. You will see a "VASCO_FM_THALES" listing in the **Value** column if your license contains this feature. Otherwise, request a new license.



Note: The VASCO_FM_THALES vendor must be one of valid vendors within the system.

Creating Token Policy

Administrators can configure nCipher HSM characteristics by creating a new Token Policy via the AccessMatrix Admin Console.

1. Perform the steps described in [Navigating Token Policy](#).
2. Click the **Create** button. The **Choose Token Policy Implementation** page is displayed.
3. Select "VASCO Kernel Parameter (Thales SEE Enabled)" from the **Implementation** list. Then, click the **Next** button. The **Create Token Policy** is displayed.
4. Specify the desired **Token Policy Id** and **Token Policy Name**.
5. Click **Save** to commit the changes.



Notes:

- The upper section of the page contains common attributes such as: **Token Policy Id**, **Token Policy Name**, **Description**, **Implementation**, and **Encryption key**.
 - The lower section of the page contains the **Attributes** tab.
 - "*" denotes a mandatory attribute.
-

Attribute Name	Description
Attributes Tab	
Thales Specific	
*Storage Key Name	Refers to the VASCO storage key name.
VASCO Specific	
Storage Derive Key 1	Derivation key part 1 used to make data software encryption unique for a host.
Storage Derive Key 2	Derivation key part 2 used to make data software encryption unique for a host.
Storage Derive Key 3	Derivation key part 3 used to make data software encryption unique for a host.
Storage Derive Key 4	Derivation key part 4 used to make data software encryption unique for a host.
Derive Vector	The vector used to make data encryption unique for a host. Default: 0
Diag Level	The level of diagnostic messages generated by functions, which goes to standard output. Default: 0
Check Challenge	Verify or not if the challenge has been corrupted before validation. Options: "0" = no password checking "1" = check parameter then verify with the DPData Challenge "2" = always use the DPData Challenge to validate responses "3" = avoid challenge-response replay attack by allowing only one challenge-response authentication per time step "4" = avoid challenge-response replay attack by rejecting the second response if responses from 2 consecutive authentication requests are equal and in the same time step. Default: 0
Check Inactive Days	Acceptable number of days of user/token inactivity. Past this number return code 205 will be generated and the Digipass needs to be reset. "0" means feature is disabled. Value must be between 0 to 1024. Default: 0
Event Window	Event Window size, expressed in number of iterations. Represents the acceptable event counter difference between digipass and host. Applicable only

Attribute Name	Description
	for event-based operating modes. Value must be between 10 to 1000. Default: 100
GMTAdjust	Number of secs – GMT time adjustment to perform in case the C language gmtime function doesn't give an accurate value. Value must be between -86400 to 86400. Default: 0
IThreshold	Number of successive Identification errors allowed. When the specified number is reached, return code 202 is sent to caller. The token will then need to be re-synced. Value must be between 0 to 255. Default: 3
ITime Window	The number of timesteps that determines the acceptable time difference between a Digipass and the host system for Identification function. This difference is adjusted to the last known shift for each token. The max drift correction acceptance is 1 sec per 6 hr period. The time step is determined by the Digipass internal programming options. Please refer to the concepts pages to have more information concerning the Time Management in the Vacman controller. Value should not be less than 2. Default: 10
ITime Window 1	Identification timesteps value 1.
ITime Window 2	Identification timesteps value 2.
ITime Window 3	Identification timesteps value 3.
Online SG	Level of Online Signature. Options: • "0" = disabled feature • "1" = several signatures are allowed in same timestep (except identical successive ones) • "2" = only one signature per timestep allowed Default: 2
SThreshold	Number of successive Signature errors allowed. When the specified number is reached, the return code 203 is sent to caller. The token will then need to be re-synced. Value must be between 0 to 255. Default: 3
STimeWindow	The number of timesteps that determines the acceptable time difference between a Digipass token and the host system for Signature function. This difference is adjusted to the last known shift for each token. The time step is determined by the Digipass internal programming options.

Attribute Name	Description
	Value should not be less than 2. Default: 10
STimeWindow 1	Signature timesteps value 1.
STimeWindow 2	Signature timesteps value 2.
STimeWindow 3	Signature timesteps value 3.
Synchronize Window	For InitialTimeWindow Configuration. This parameter sets the initial derivation between a Digipass time and the Vacman Controller GMT time. This value is expressed in hours. InitTimeWindow = max(STimeWindow or ITimeWindow), (SyncWindow). Value must be between 1 to 60. Default: 2
Synchronize Window 1	InitialTimeWindow configuration 1.
Synchronize Window 2	InitialTimeWindow configuration 2.
Synchronize Window 3	InitialTimeWindow configuration 3.
Preferred application ID for OTP (RO) if more than one blob	Specify preferred application Id for verification if token has more than one RO blobs.
Secure Channel Application ID(s)	Specify the application Id(s) for Secure Channel application that matches the application Id(s) defined on VASCO's parameter sheet.
Secure Channel Message Validity Time	This value indicates the Secure Channel(SC) Message validity time in seconds. During Secure Channel verification, this validity time is used to check whether the generated Secure Channel message has expired. The default value "0" means that this is disabled. Default: 0
VASCO Double DPX	Enable import for double encrypted DPX files. Default: false
Digipass for APPS/Mobile Specific Tab	
Digipass for APPS/Mobile Specific	

Attribute Name	Description
Server Activation Challenge Response	Enables challenge response verification for activation process if set to "true". Default: false
Score Based Response	Enables Score Based Response (contextual) verification if set to "true". This feature will perform verification on OTP response which contains User, Context and Platform scoring generated by user's VASCO token. Default: false
Generate and Send Mobile Activation QRCode After Assignment	This feature is strictly for Mobile Tokens only. If enabled, server will generate Mobile Activation QRCode upon successful token assignment. This Activation QRCode will be sent to user through event notification policy. Please select corresponding event notification policy in Event Notification Policy for Activation QRCode attribute. Default: false
Event Notification Policy for Mobile Activation QRCode	Specify the event notification policy to be used for sending the Mobile Activation QRCode.
Allow Activation Checking	If set to "true", special validation will be performed during Digipass Mobile token activation to check if the token's 'Allow Activation/Reactivation' flag is set to "YES". If token's 'Allow Activation/Reactivation' flag is set to "NO", the token activation will be denied. Set to "false" to disable this validation. Default: false
Digipass Online Activation Secured Data Exchange Protocol	
Secured Data Exchange Protocol	Specify the type of protocol to be used for Secured Data Exchange during Digipass for APPS/Mobile online activation. Options: "SRP" - Secure Remote Password protocol "ECDH" - Elliptic curve Diffie-Hellman protocol Default: ECDH
SRP Activation Specific	
User Password Length	SRP's user password length. Default: 8
User Password Character Sets	SRP's user password character sets. Options: "Numeric" "Alphanumeric" Default: Numeric
User Password Letter Case	SRP's user password letter case. Options: "Lowercase Alphanumeric" "Uppercase Alphanumeric"

Attribute Name	Description
	"Mixed Case Alphanumeric" Default: Mixed Case Alphanumeric
Checksum on User Password	If set to "true", VASCO SDK will calculate the checksum of the user's password. Default: true
User Password Validity Time (Seconds)	This value indicates the SRP Password validity time in seconds. During password verification, this validity time is used to check whether the generated password has expired. Default: 86400
ECDH Activation Specific	
Activation Password Length	ECDH's activation password length. Default: 6
Activation Password Character Set	ECDH's activation password character set. Options: "Numeric" "Lowercase Alphabetic" "Uppercase Alphabetic" "Mixed Case Alphabetic" "Uppercase Alphanumeric" "Mixed Case Alphanumeric" "Lowercase Alphanumeric" Default: Numeric
Activation Password Validity Time (Seconds)	This value indicates the Activation Password validity time in seconds. During password verification, this validity time is used to check whether the generated password has expired. Default: 86400
Authorization Code	
Authorization Code Length	Refers to the authorization code's length requirement. Default: 6
Authorization Code Character Set	Refers to the authorization code's character sets requirement. Default: Numeric
Authorization Code Validity Time (Seconds)	This value indicates the Authorization Code validity time in seconds. During authorization code verification, this validity time is used to check whether the generated authorization code has expired. Default: 86400

Attribute Name	Description
Authorization Code Retry Threshold	The maximum failed attempts for Authorization Code verification during Activation. The Authorization Code will be invalidated upon threshold reached. Set to "0" to disable this feature. Default: 3
CrontoSign Configuration	
Use Dynamic CrontoSign	Enforce the use of dynamic size CrontoSign image generation to support up to 1070 characters for Secure Channel transaction signing data. If disabled, standard size CrontoSign(support up to 200 characters) will be used instead. Default: false

Creating Token Group

To create a token group for OneSpan SEE token:

1. Perform the steps described in [Navigating Token Group](#).
2. Click the **Create** button. The **Create Token Group** page is displayed.
3. Specify the desired **Token Group Id** and **Token Group Name**.
4. Select "VASCO_FM_THALES" as the **Token Vendor** for this token group.
5. Click **Save** to commit the changes.

Creating Token Batch

You are required to use a token policy with VASCO Kernel Parameter (Thales HSM SEE Enabled) implementation for your token batch. To create a token policy for that, refer to the [Creating Token Policy](#) section.

To create Token Batch:

1. Perform the steps described in [Navigating Token Batch](#).
2. Click the **Create** button. The **Create Token Batch** page is displayed.
3. Supply the desired **Token Batch Id** and **Token Batch Name**. Select the **Token Policy** value created in previous steps.
4. Click **Save** to commit the changes.

Validating/Importing Entrust OneSpan Key Using TIU

To validate existing OneSpan (formerly VASCO) keys in Entrust OneSpan Key file path or import a new OneSpan key from Entrust OneSpan Key file path, you need to make sure:

- The Entrust OneSpan Application key files path is configured in **amsystem.properties**. This path refers to %NFAST_KMDATA%\Key Management Data\, the path that contains all the generated OneSpan Cryptographic keys from the section [Managing OneSpan Keys](#).
- AccessMatrix server has VASCO_FM_THALES token vendor in its server license. Refer to the [Validating Token Vendor](#) section.

1. Login to TIU by following the steps described in [Launching TIU](#).
2. Click on the **Thales Vasco Key Import** button. The OneSpan keys table is displayed. Each column is explained below:

- **Key Label:** Vasco key type that is auto-labeled by Key Manager Tool after the key is generated.
For example: vascStorageKey, vascTransportKey, vasc1, vasc2... etc.
- **Key Index:** Vasco key ID.
- **Uploaded Date:** Last update date of the key from the file path to the database.
- **sha1sum of key in DB:** sha1sum of the key in database. Compares the integrity of the key with the key inside the file path.
- **Key File Name:** Vasco key file name that is auto-named by the Key Manager Tool after the key is generated.

 **Note:** Do not modify the filename.

- **sha1sum of Key in File:** sha1sum of the key in file path. To compare integrity of the key with the key inside database.
 - **sha1sum Comparison (DB vs File):** Compares the keys between the database and file. If it is the same key, the value is "true"; otherwise, it is "false".
 - **Load to DB:** Button to load the key from file path into database.
 - **Clear value in DB:** Button to clear the key inside the database.
3. If the key already exists in the file path, you will see the value in **sha1sum of Key in File**.
 4. Click the **Load** button to upload the key into the database.

-
5. After the key is uploaded, both the **sha1sum of Key in File** and **sha1sum of key in DB** will share the same value. In the **sha1sum Comparison (DB vs File)** column, the value will be displayed as "true". An upload date and time will also be displayed in **Uploaded Date** column as well.
 6. Try to upload the rest of the keys.
 - If the key is regenerated by the Key Manager Tool, the keys in the file and database will be different. You will see different values for **sha1sum of Key in File** and **sha1sum of key in DB**. In the **sha1sum Comparison (DB vs File)** column, the value will be "false".
 7. Click the **Clear** button to clear the key inside database.
 8. Click the **Load** button to upload new key from file path into database.

Importing OneSpan SEE Token Using TIU



Notes: If you are creating a token batch and token group in TIU, ensure the following:

- Your AM Server license must support VASCO_FM_THALES token vendor.
- Use the token policy created in [Creating Token Policy](#) for your token batch. To create a token batch, read [Creating Token Batch](#).
- Use VASCO_FM_THALES as the token vendor for your token group. To create a token group, read [Create Token Group](#).

Importing Single Encryption DPX

1. Login to TIU by following the steps described in [Launching TIU](#).
2. Click on the **Import Token** button.
3. Upload the DPX file and specify the token batch with the VASCO SEE enabled token policy, and the token group with the VASCO SEE vendor present.
4. Enter the **Transport Key** provided by VASCO.
5. Click **Preview Token** to load the token. The list of token serial number will be displayed if the VASCO SEE token is imported successfully.
6. Click **Import Token** to import the tokens.

Importing Double Encryption DPX

Prior to importing double encryption dpx file, you need to increase the key component value of the token import. To perform the configuration:

-
1. Perform the steps described in [Navigating Token Management](#).
 2. Modify the **Token Import - Number of key components** attribute (the value depends on number of key components provide by VASCO).
 3. Click **Save** to commit the changes.
 4. Perform the steps described in [Navigating Token Policy](#).
 5. Select the token policy created in previous steps.
 6. Click the **Edit** button and change the **VASCO Double DPX** value to "true".
 7. Click **Save** to commit the changes. Then, log in to TIU for double DPX login.
 8. Login to TIU by following the steps described in [Launching TIU](#).
 9. Click on **Import Token**.

i **Note:** The token import instructions are similar to token import for single encryption DPX file as described in [Importing Single Encryption DPX](#). The difference is that this method requires two software-level DPX transport keys.

Assigning Token to a User

In order to ensure that the OneSpan SEE token is supported, administrator is required to assign the token to the user.

- i** **Notes:**
1. Make sure Authentication Realm (e.g. "40192") is pointing to the targeted user store. To navigate and configure Authentication Realm, refer to [AccessMatrix Common Administration Guide - Configure Authentication Realm](#).
 2. Make sure the Login Module token vendor name is correctly pointing to VASCO_FM_THALES. To navigate and configure Login Module, refer to [AccessMatrix Common Administration Guide - Configure Login Module](#).
 3. Synchronize the token before assigning to user.
 4. Assign the token and give the user the correct authentication realm.

To test the OneSpan SEE login, administrator may use the `usoagent` or API to test the login.

4. Appendices

This section contains the following appendices:

- [A. Common Admin Console Configuration Steps](#)
- [B. Customer's Key Attributes Permission](#)
- [C. Testing Using VACMAN Controller](#)

4.1. A. Common Admin Console Configuration Steps

Loading Administration Dashboard

1. Log in to the Admin Console.
2. Select **Administration Dashboard** from the **Dashboard** drop-down list found at the upper right corner of the first-level navigation bar.

Loading Operation Dashboard

1. Log in to the Admin Console.
2. Select **Operation Dashboard** from the **Dashboard** drop-down list found at the upper right corner of the first-level navigation bar.

Navigating Server License

1. Perform the steps described in [Loading Administration Dashboard](#).
2. Click **System** from the first-level navigation bar.
3. Click **Server & Cache** from the second-level navigation bar.
4. On the left pane, click **Server** and then click the **Create** button. The **Search Server** form and **Search Server Result** will be displayed.
5. Click **Search** and select the Server Id of your server.
6. Click the **Licensed Features** tab.

Navigating Token Policy

1. Perform the steps described in [Loading Administration Dashboard](#).
2. Select **Security Policy** on the first-level navigation bar.
3. Choose the **Token** option on the second-level navigation bar.
4. On the left pane, select the **Token Policy** option, then click the **Create** button. The **Search Token Policy** form and **Search Token Policy Result** will be displayed.

Navigating Token Group

1. Perform the steps described in [Loading Operation Dashboard](#).
-

-
2. Select **Token** on the first-level navigation bar.
 3. Choose the **Token Group** option on the second-level navigation bar.
 4. On the right pane click the **Create** button. The **Search Token Group** form and **Search Token Group Result** will be displayed.

Navigating Token Batch

1. Perform the steps described in [Loading Operation Dashboard](#).
2. Select **Token** on the first-level navigation bar.
3. Choose the **Token Batch** option on the second-level navigation bar.
4. On the right pane, click the **Create** button. The **Search Token Batch** form and **Search Token Batch Result** will be displayed.

Launching TIU

1. Launch the AccessMatrix Admin Console by accessing the AM5 URL, e.g.
http://localhost:8080/am5.
2. Click the **Cancel** button to remove the current login screen of the Admin Console.
3. At the upper left corner of the page, click **Token > Token Import Utility**.
4. Enter **Login Id** and **Password for Login** (User 1), e.g. "amadmin" / "password". Then, click the **OK** button.
5. Enter **Login Id** and **Password for Login** (User 2), e.g. "amadmin1" / "password". Then, click the **OK** button.
6. The system will display the **Token Import Utility (TIU)** page.

Navigating Token Management

1. Perform the steps described in [Loading Administration Dashboard](#).
 2. Select **Configuration** on the first-level navigation bar.
 3. Select the **Global Setting** option on the second-level navigation bar. On the right pane, the **View Global Setting** page will be displayed.
 4. Select the **Token Management** tab.
-

4.2. B. Customer's Key Attributes Permission

Expected Values

1. The table below contains the expected attribute values for the keys on the customer's side:

Key Attributes	HSM Storage Key	HSM Transport Key	KEK
Key Type	DES2 or DES3	DES2 or DES3	DES2 or DES3
Key Size	128 or 192	128 or 192	128 or 192
Key Module Protected (1)	TRUE	TRUE	TRUE
Key SEE Protected (2)	TRUE	TRUE	TRUE
Encrypt Operational Permission	TRUE(3)	TRUE(3)	FALSE(4)
Decrypt Operational Permission	TRUE(3)	TRUE(3)	FALSE(4)
GetACL Operational Permission	TRUE	TRUE	TRUE
Export	FALSE	FALSE	TRUE(5)
Exportable	FALSE	TRUE(6)	FALSE

- a. All the keys created by the manager tool are protected by the Security World. The manager creates keys in standalone files (up to 11 key files: **vascoStorageKey.txt**, **vascoTransportKey.txt**, and **vasco1.txt** to **vasco9.txt**) containing the key blobs protected by the module (Security World Key).
 - b. All the keys created by the manager tool have their operational permissions protected by the SEE signing certificate used to sign the VACMAN Controller SEE machine.
 - c. HSM-level BLOB storage keys and HSM-level DPX transport keys will be allowed to perform encrypt and decrypt operations only within the VACMAN Controller SEE machine.
 - d. Key Encrypting Keys will be allowed to perform neither encrypt nor decrypt operations. These keys will be used outside the VACMAN Controller SEE machine context, to export one or more HSM-level DPX transport key(s) in encrypted form (See step e).
-

-
- e. Generated Key Encrypting Keys will be able to export in encrypted form only keys that will be defined as exportable by this KEK (HSM-level DPX transport keys).
 - f. Generated HSM-level DPX transport keys will be exportable with only one defined KEK (the KEK prone to export the HSM-level DPX transport key chosen during transport key generation when using the manager tool).

2. The table below contains the meaning of the various Key attributes:

Attribute	Meaning
Key Module Protected	Indicates whether the key is protected by the Security World Key. The key will not be usable in another Security World.
Key SEE Protected	Indicates whether the key operational permissions (Encrypt, Decrypt, GetACL) are protected by the SEE signing certificate used to sign the VACMAN Controller SEE machine. The key will not allow performing these operations (if permitted) outside the VACMAN Controller SEE machine, or into another SEE machine.
Encrypt Operational Permission	Indicates whether the key can be used to encrypt data. This operation will be possible only within the VACMAN Controller SEE machine.
Decrypt Operational Permission	Indicates whether the key can be used to decrypt data. This operation will be possible only within the VACMAN Controller SEE machine.
GetACL Operational Permission	Indicates whether the key can be used to list all the permitted operations (Encrypt, Decrypt, GetACL). This operation will be possible only within the VACMAN Controller SEE machine.
Export	Indicates whether the key can be used to export an exportable key in encrypted form. The key will be authorized to act as a Wrapping Key in a Raw Encrypt Derivation Mechanism.
Exportable	Indicates whether the key can be exported in encrypted form. The key will be authorized to act as a Base Key in a Raw Encrypt Derivation Mechanism using a specified key as Wrapping Key.

4.3. C. Testing Using VACMAN Controller

This chapter illustrates the use of VASCO sample program for testing VASCO SEE performance. This sample program is provided by VACMAN Controller to help software developer to test the usability of VASCO SEE token.

Install VACMAN Controller nCipher HSM

i **Note:** For testing, ensure that VACMAN Controller nCipher HSM has been installed in client machine and the configurations below have been done.

1. Access the VACMAN Controller nCipher HSM folder: *<Vasco installed path>\Sample\java*.
2. Create a new folder named *dpx*. Copy **demo.dpx** and **demodblencrypt.dpx** dpx files from *<Vasco installed path>\Dpx* to the new folder.
3. Copy **aal2wrap.jar** from *<Vasco installed path>\Wrappers\java* to jar folder.
4. Modify **NcipherWrap.java** in *src* folder.
 - a. Modify the declared SEEWorld to PublishedSEEWorld:

Java

```
...
private static NFConnection conn;
//Modify SEEWorld with PublishedSEEWorld
/*private static SEEWorld sw = null;*/
private static PublishedSEEWorld sw = null;
private static SecurityWorld secworld = null;
...
```

- b. Modify the instantiated SEEWorld to PublishedSEEWorld:

i **Note:** The "vascoSEE" below is the dummy data in userdata.txt - please specify your dummy data.

Java

```
...
try
{
    // Create the SEEWorld using the constructor
    (connection, ModuleID, SAR name,
    // debug enabled, certificate chain to allow debug)
    String userDataFile =
    System.getProperty("KMDATA");
    userDataFile += mysarfilename;
    //Modify SEEWorld with PublishedSEEWorld
    //sw = new SEEWorld(conn, MODULEID, mysarfilename,
```

```

false, null);
        sw =new
PublishedSEWorld(secworld.getConnection(),
MODULEID, "vascoSEE");
    }
    catch(Exception e)
    {
        //Remove 2nd call of SEWorld
        /*try
        {
            // If 1st call of SEWorld failed, is maybe
            due to version of codesafe;
            // in this case call SEWorld with only
            mysarfilename as input
            sw = new SEWorld(conn, MODULEID,
            mysarfilename, false, null);
        }
        catch (Exception e2)
        {
            System.err.println("Could not create
SEWorld:");
            e2.printStackTrace();*/
            e.printStackTrace();
            CloseConnection(secworld.getConnection());
            System.exit(1);

        /*}*/

    }
    ...

```

5. Modify following batch files **buildtest.bat** in the Java folder:

Example:

```

javac -d ./class -sourcepath ./src -classpath
"%NFAST_HOME%/java/classes";"%NFAST_HOME%/java/classes/nfjava.jar";"%NFAST_HOME%/java/c
lasses/kmjava.jar";"%NFAST_HOME%/java/classes/jhsee.jar" src/NcipherWrap.java
javac -d ./class -sourcepath ./src -classpath
"%NFAST_HOME%/java/classes";jar/aal2wrap.jar;class;NcipherWrap.class src/aal2tst.java

```



Note: Make sure the classpath is pointing correctly to the desired jar location. Take note that different machine might have a different location setting. Please modify accordingly if they are different from the example given.

Testing Single Encryption

1. Open a command prompt window with administrator rights.
2. Browse to the path *.../sample/java* that stores the batch files created in previous section.
3. Run the **buildtest.bat**.
4. If the java classpath directories are set correctly, the command will not return any error.
5. Next is to run the **runtest.bat**.

6. Within the same directory, run **runtest.bat**. If there is not any error in the java classpath directory, following printout will be shown:

```
Administrator: C:\Windows\System32\cmd.exe - runtest.bat

C:\Program Files (x86)\VASCO\UACMAN Controller nCipher HSM 3.14.0\Sample\java>runtest.bat
C:\Program Files (x86)\VASCO\UACMAN Controller nCipher HSM 3.14.0\Sample\java>echo off
C:\Program Files (x86)\VASCO\UACMAN Controller nCipher HSM 3.14.0\Sample\java>java -DKMDATA="C:\ProgramData\nCipher\Key Management Data/" -classpath "C:\Program Files (x86)\nCipher\nfast\java\classes";"C:\Program Files (x86)\nCipher\nfast\java\classes\nfjava.jar";"C:\Program Files (x86)\nCipher\nfast\java\classes\kmjava.jar";"C:\Program Files (x86)\nCipher\nfast\java\classes\jhsee.jar";class;.jar /aal2wrap.jar aal2tst
*****
UACMAN Controller for nCipher HSM test program

You need a VASCO demo token.
The demo DPX file have to be located in the %cwd%\dpx
directory.
You are running the UC with an nCipher HSM.

You need to create the vascoStorageKey in the HSM
You need to import the correct vascoTransportKey in the HSM
(Transport Key KCU = 5734E3)

To import this Transport Key, please read the
Keys Dbl Encrypt Demo.txt file provided in the dpx directory and the
UACMAN Controller Key Management For nCipher netHSM.pdf in the
documentation directory
*****
Time settings :
Local Time : 30 March 2015 18:11:33 SGT
Gmt Time : 30 March 2015 10:11:33 GMT

press enter to continue...

UC Test 1 AAL2DPXGetTokenBlobsEx DPX Import Demo Token - Single Encryption
UC Test 1 AAL2DPXGetTokenBlobsEx DPX Import Demo Token - Single Encryption : OK

SecurityWorld Created:
Module find
SEESWorld Created:
confKeyHandle = 1773881406
Key loaded - vascoStorageKey.txt
Ticket created
Ticket Sent
confKeyHandle = 1773881407
Key loaded - vascoTransportKey.txt
Ticket created
Ticket Sent

UC Test 2 AAL2ProcMigrateBlobReply Migration Demo Appl 1 - Single Encryption to Storage Key
UC Test 2 AAL2ProcMigrateBlobReply Migration Demo Appl 1 - Single Encryption to Storage Key : OK

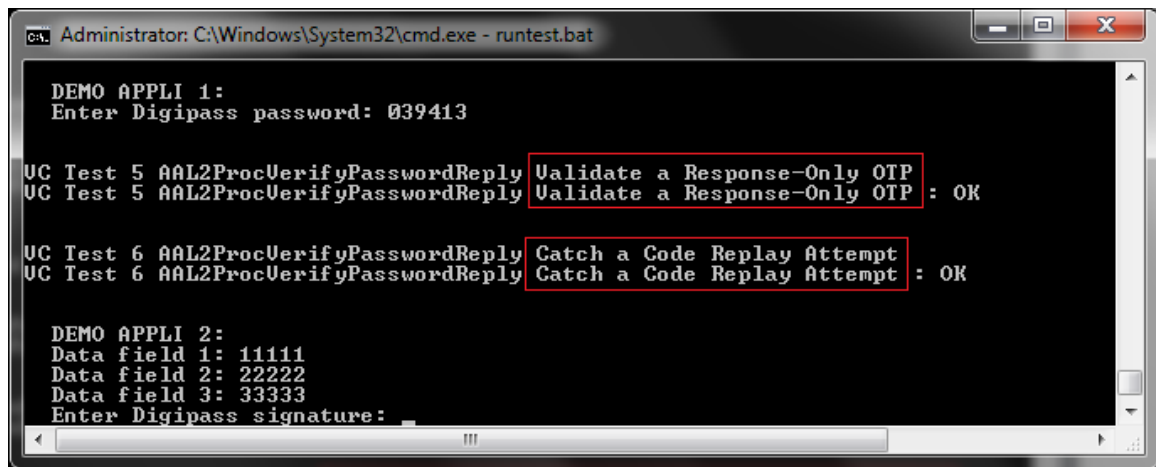
UC Test 3 AAL2ProcMigrateBlobReply Migration Demo Appl 2 - Single Encryption to Storage Key
UC Test 3 AAL2ProcMigrateBlobReply Migration Demo Appl 2 - Single Encryption to Storage Key : OK

UC Test 4 AAL2ProcMigrateBlobReply Migration Demo Appl 3 - Single Encryption to Storage Key
UC Test 4 AAL2ProcMigrateBlobReply Migration Demo Appl 3 - Single Encryption to Storage Key : OK

DEMO APPLI 1:
Enter Digipass password: _
```

7. For the Test 1 - Test 4 printouts, the administrator needs to ensure that the result is Single Encryption to Storage Key: OK for DEMO APPLI 1, 2, and 3.
8. Testing on single encryption, administrator is required to enter the Digipass password generated by any sample token for DEMO APPLI 1. Otherwise, press Enter to skip this single encryption input.
9. Continue with the single encryption verification by entering the Digipass password. The following printouts should be displayed:
 - a. **Test 5** - Validate a Response-Only OTP: OK
 - b. **Test 6** - Catch a Code Replay Attempt: OK

-
10. Continue to enter the Digipass signature for DEMO APPLI 2. Please read [Testing Digital Signature](#).



```
Administrator: C:\Windows\System32\cmd.exe - runtest.bat

DEMO APPLI 1:
Enter Digipass password: 039413

UC Test 5 AAL2ProcVerifyPasswordReply Validate a Response-Only OTP : OK
UC Test 5 AAL2ProcVerifyPasswordReply Validate a Response-Only OTP : OK

UC Test 6 AAL2ProcVerifyPasswordReply Catch a Code Replay Attempt : OK
UC Test 6 AAL2ProcVerifyPasswordReply Catch a Code Replay Attempt : OK

DEMO APPLI 2:
Data field 1: 11111
Data field 2: 22222
Data field 3: 33333
Enter Digipass signature: _
```

Testing Digital Signature

1. Testing on digital signature, administrator is required to enter the Digipass signature generated by any sample token with Digipass Signature for DEMO APPLI 2 and DEMO APPLI 3. Otherwise, press Enter to skip this single encryption input.
2. Continue with the single encryption verification by entering the Digipass signature. The following printouts should be displayed:
 - a. **Test 7** - Validate a signature: OK
 - b. **Test 8** - Validate a Challenge/Response OTP: OK
3. Make sure Test 9 and Test 10 have the result Double Encryption: OK / Double Encryption Transport Key to Storage Key: OK.

-
4. Continue to enter the Digipass password for Double Encryption DEMO APPLI 1. Please read [Testing Double Encryption](#).

```
Administrator: C:\Windows\System32\cmd.exe - runtest.bat

DEMO APPLI 2:
Data field 1: 11111
Data field 2: 22222
Data field 3: 33333
Enter Digipass signature: 717734

UC Test 7 AAL2ProcVerifySignatureReply Validate a signature
UC Test 7 AAL2ProcVerifySignatureReply Validate a signature : OK

DEMO APPLI 3:
Enter Digipass password for challenge 5888:0638366

UC Test 8 AAL2ProcVerifyPasswordReply Validate a Challenge/Response OTP
UC Test 8 AAL2ProcVerifyPasswordReply Validate a Challenge/Response OTP : OK

UC Test 9 AAL2DPXGetTokenBlobsEx DPX Import Demo Token - Double Encryption
UC Test 9 AAL2DPXGetTokenBlobsEx DPX Import Demo Token - Double Encryption : OK

UC Test 10 AAL2ProcMigrateBlobReply Migration - Double Encryption Transport Key to Storage Key
UC Test 10 AAL2ProcMigrateBlobReply Migration - Double Encryption Transport Key to Storage Key : OK

DEMO APPLI 1:
Enter Digipass password: _
```

Testing Double Encryption

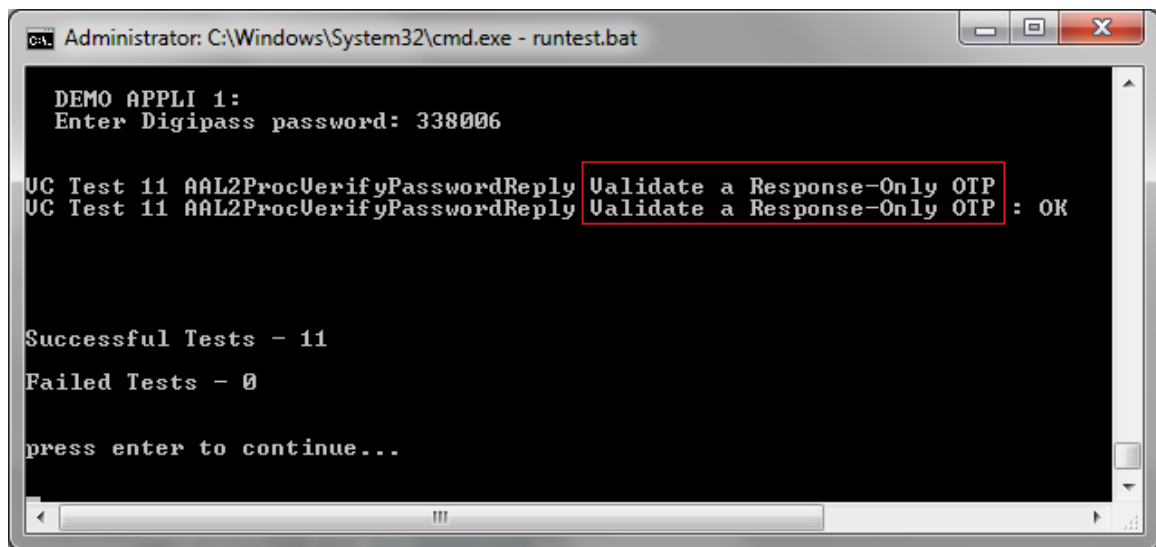


Note: Ensure the following prerequisites are met:

- The dpx file used for this testing is called **demodblencrypt.dpx** and it should be located at the *dpx* folder.
- Ensure that the transport key has been imported, and unwrapped by the KEK key. The transport Key KCV is 5734e3.

1. For the testing of double encryption, the administrator is required to enter the Digipass password generated by any sample token for DEMO APPLI 1. Otherwise, press Enter to skip this double encryption input.
 2. The Test 11 printout should show the result Validate a Response-Only OTP: OK.
 3. The number of Successful Tests and Failed Tests is also displayed.
-

-
4. Press Enter to exit the test program.



```
Administrator: C:\Windows\System32\cmd.exe - runtest.bat

DEMO APPLI 1:
Enter Digipass password: 338006

UC Test 11 AAL2ProcVerifyPasswordReply Validate a Response-Only OTP
UC Test 11 AAL2ProcVerifyPasswordReply Validate a Response-Only OTP : OK

Successful Tests - 11
Failed Tests - 0

press enter to continue...
```
